MATH317 · NUMERICAL METHODS

Midterm 462 Solutions

18/03/2025 · 60 minutes · 30 marks

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01 Solution Q1: errors and Horner

1a) Absolute error = $|\ 30.158 - 30.16\ | = 0.002$

1b) Relative error = $0.002 / 30.158 = 0.66 \times 10^{-4}$

$= 0.0066\ \%$

2a)

	4	−8	14	−32	
3 |		12	12	78	
	4	4	26	46	

$P(3) = 46$

2b)

	4	−8	14	−32	
2 |		8	0	28	
	4	0	14 |	−4	$c0 = -4$
		8	16		
	4	8 |	30		$c1 = 30$
		8			
	4 |	16			$c2 = 16$
	4				$c3 = 4$

$P(x) = 4(x-2)^3 + 16(x-2)^2 + 30(x-2) - 4$

02 Solution Q2: double precision

1. $(184)_{10} = (270)_8 = (1011\ 1000)_2$

2. $(.125)_{10} = (.1)_8 = (.001)_2$

3. $(184.125)_{10} = (1011\ 1000.001)_2 = (1.011\ 1000001)_2 \times 2^7$

   stored mantissa: .011 1000001

4. $c - 1023 = 7 \quad \rightarrow \quad c = 1030 = (2006)_8 = (100\ 0000\ 0110)_2$

5. $[1 \mid 10000000110 \mid 0111000001\ 000...0]_2$

   $[1100\ 0000\ 0110\ 0111\ 0000\ 0100\ 0000\ ...\ 0000]_2$

   $= [C067040000000000]_{16}$

Notes on the key

The key writes the answer as `[C0670400]16` without the trailing zeros. A full double-precision answer has 16 hex digits.

Step 3 of the key shows `1259.125` and step 4 shows `130`. Both are typos; the correct values are `184.125` and `1030`.

03 Solution Q3: Bisection

Part 1: number of iterations

$$n > [\ \log(b - a) - \log(2\varepsilon)\ ] / \log 2$$

$$n > [\ \log(1) - \log(2 \times 10^{-2})\ ] / \log 2 = 5.64$$

The smallest integer that works is `n = 6`. Since counting starts at `n = 0`, that is `7` iterations. The key writes: `5.6 ≈ 7`.

Part 2: finding the root

```
u = f(0) = −1          v = f(2) = 8 − e² = 8 − 7.3891 = 0.6109
```

```
u · v < 0   →   f has a root in [0, 2]
```

```
n = 0:  c = (0 + 2)/2 = 1          w = f(1) = −0.71828
```

```
        u · w > 0   →   a = c   →   [1, 2]
```

```
n = 1:  c = (1 + 2)/2 = 1.5        w = f(1.5) = 0.01831
```

```
        u · w < 0   →   b = c   →   [1, 1.5]
```

```
n = 2:  c = (1 + 1.5)/2 = 1.25     w = f(1.25) = −0.36534
```

```
        u · w > 0   →   a = c   →   [1.25, 1.5]
```

n	c_n	$f(c_n)$	Error
0	1	−0.71828	1
1	1.5	0.01831	0.5
2	1.25	−0.36534	0.25

At `n = 1` the sign changes: `u` is now the negative `f(1)` and `w` is positive, so the root is in the left half.